KnowGenie AI: Intelligent Document Retrieval & Summarization System

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Abstract

KnowGenie AI is an intelligent system that retrieves relevant information from documents and generates summaries using a Retrieval-Augmented Generation (RAG) pipeline. The system leverages a modern web frontend built in React, a robust backend powered by Express.js and Node.js, and Google Gemini API for AI-driven summarization, FAQ generation, and chat responses. By combining vector search with LLM capabilities, KnowGenie AI enables fast, accurate, and context-aware document interaction.

Introduction

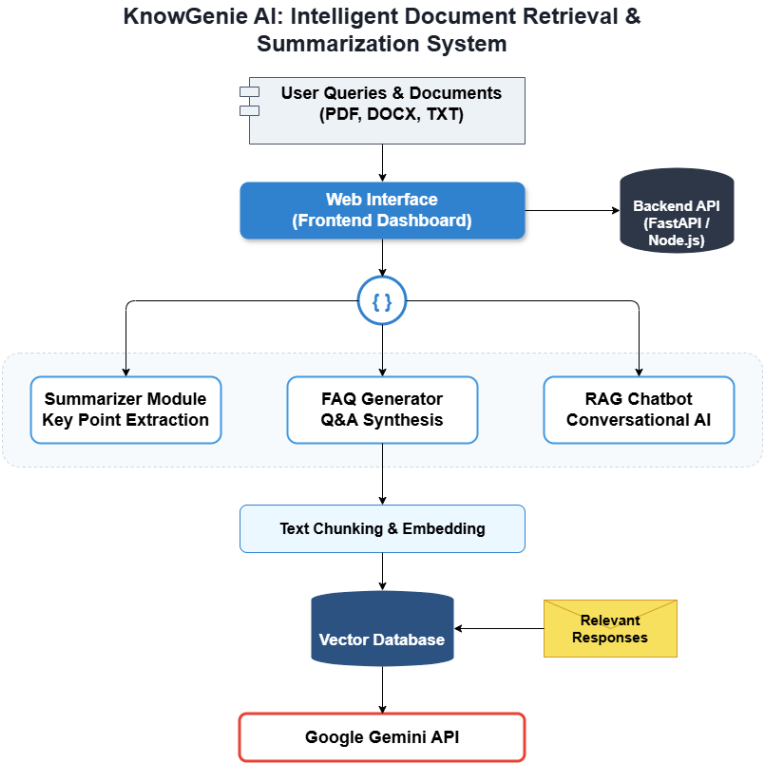
- Motivation: In the era of information overload, extracting meaningful insights from large document collections is challenging. KnowGenie AI addresses this by providing intelligent retrieval and summarization.
- Objectives:
    - Efficient document retrieval
    - Context-aware summarization
    - FAQ generation and chat responses

System Architecture

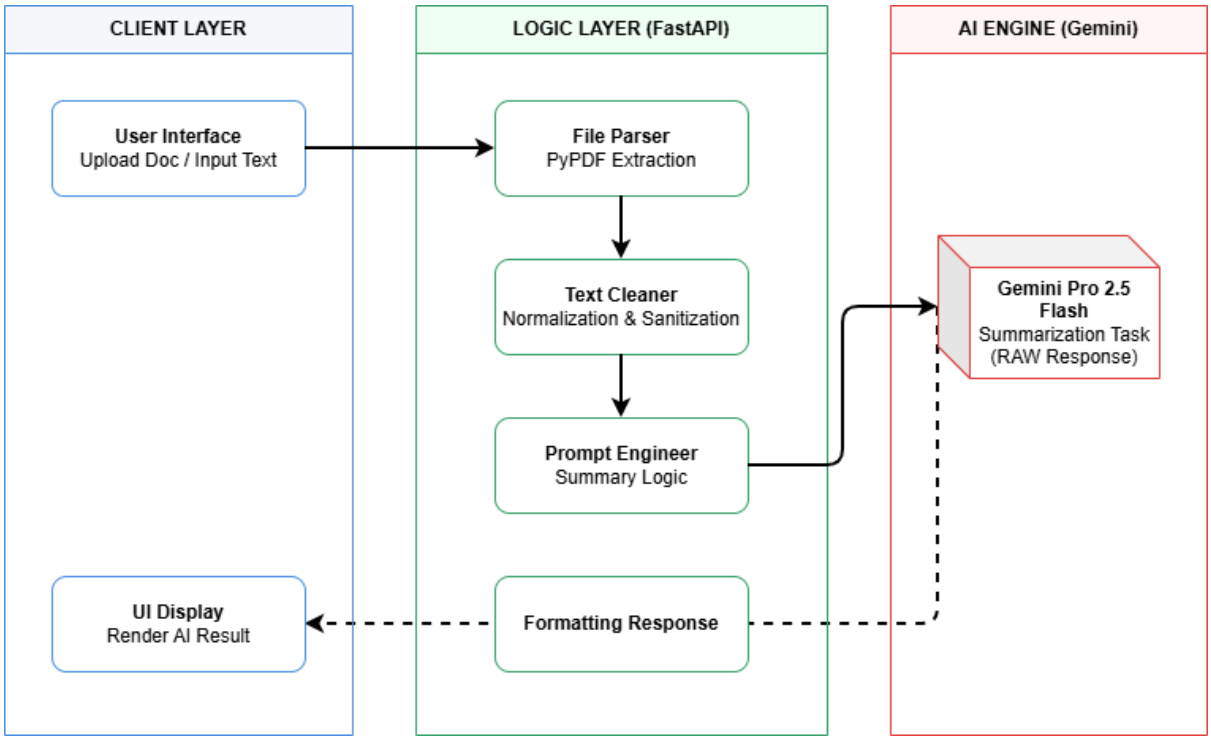
Diagram placeholder: *Draw.io*
Components:

1. Frontend: React + TypeScript + Vite + Tailwind CSS
    - UI development, type safety, styling, and HTTP communication via Fetch API
2. Backend: Express.js + Node.js
    - Handles API requests, environment configuration via dotenv, and CORS security
3. AI Layer: Google Gemini API (model: gemini-2.0-flash)
    - Provides summarization, FAQ generation, and chat response generation

Architecture:



Project Workflow:



Tech Stack Details

Frontend

- React – UI development framework
- TypeScript – Type safety and maintainability
- Vite – Fast bundler & development server
- Tailwind CSS – Utility-first styling
- Fetch API – For HTTP requests

Backend (Production-ready)

- Express.js – API server
- Node.js – Runtime environment
- CORS + dotenv – Security and environment configuration

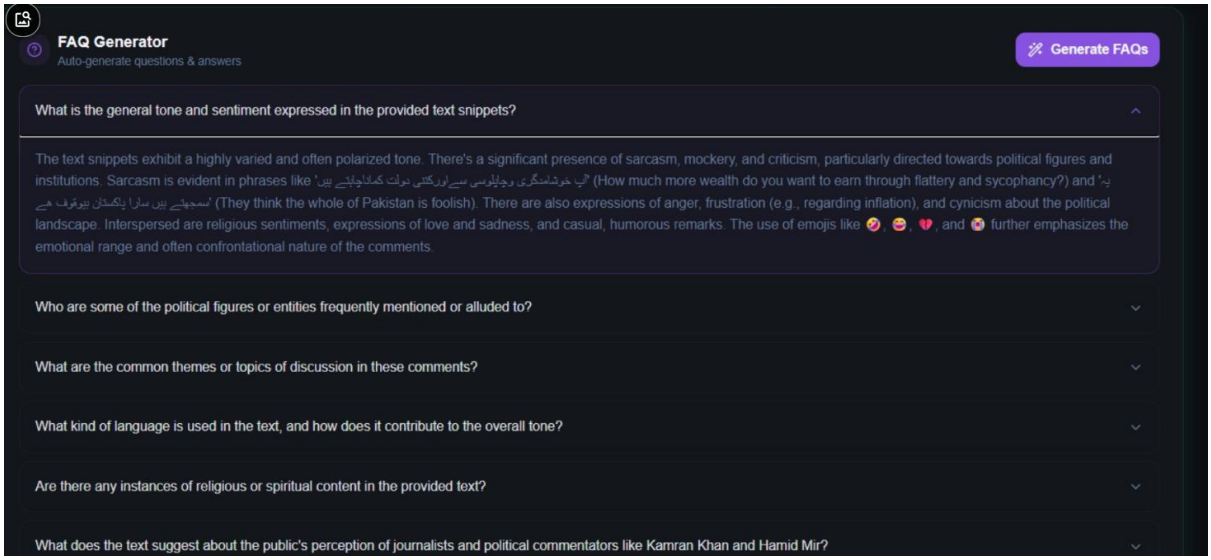
AI Layer

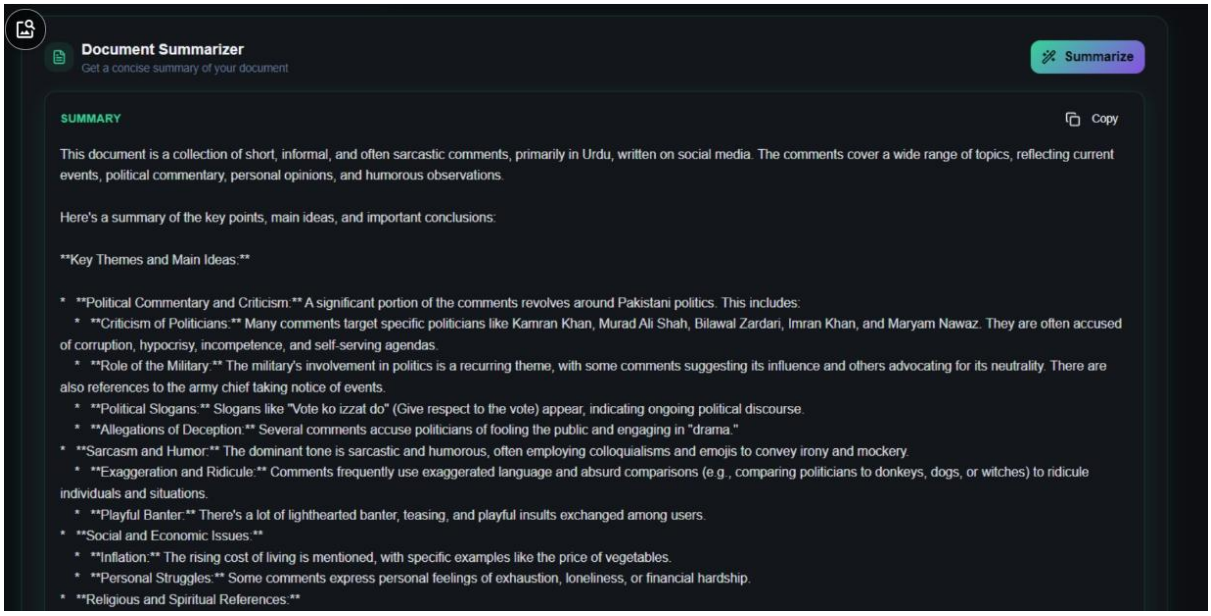
- Google Gemini API
    - Model: gemini-2.5-flash
    - Uses: Summarization, FAQ generation, Chat responses

Methodology

- Frontend:
    - Built interactive UI for query input and displaying AI responses
- Backend:
    - Secure API endpoints
    - Handles requests from frontend to AI layer
- AI Layer:
    - Retrieves relevant context using RAG pipeline
    - Gemini API generates accurate summaries, FAQs, and conversational responses

Results & Evaluation





- Performance: Fast retrieval + accurate summaries
- Optional: Compare simple LLM responses vs RAG-augmented responses

Conclusion

KnowGenie AI successfully combines modern web technologies with AI to provide an efficient and intelligent document interaction platform. Future improvements can include multi-language support and real-time streaming of documents.

References

- Vaswani et al., “Attention is All You Need”, 2017
- HuggingFace documentation for embeddings
- Google Gemini API documentation
- Tailwind CSS and React official docs